Hybrid Image Encryption and Decryption using Cryptography and Watermarking Technique for High Security Applications

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***Abstract*—Security has become a vital area of research in almost every domain. Data security has taken the front seat in many communication area of research. Everyone wants to keep their data or information safe from theft or from being corrupted. In this paper we discuss about a hybrid technique to implement data security in communication. The techniques used include watermarking and cryptography. This encryption and decryption techniques can be used for image, video, audio, and text. In this paper we have used image as the data for encryption and decryption. This paper illustrates the use of the Secure Force (SF) algorithm for encryption and decryption process. The simulation work has being carried out using MATLAB. Various parameters like PSNR, MSE and Entropy was also calculated.**

Keywords— Data Security, Hybrid, Watermarking, Cryptography, , SF-Algorithm.

I. INTRODUCTION

Today Security of Data has gained a lot of importance for all types of communication system. Extensive use of images, text had made it more significant to focus on the area of data security. While the relentless use of Internet and Communication Technology is unavoidable, in the use of data sharing, many ways to provide security to data, are also being developed to look into the issues related to hacking, unethical sharing etc.

Cryptosystem

Cryptography is a method where it converts the original data into unreadable form [1]. This technique uses two types of key to translate the data. Cryptography technique is classified into secret key cryptography and public key cryptography. Only the authorized person who has his secret key can read the unreadable form of data. Other who don't have the secret key they can't read or open the data. In private key process both the encryption and decryption keys are same. But in secret key process encryption key and decryption key are different. Here only the encryption key is made as public hence the name public key while the decryption key is not made public hence known as private key [2]. Public key cryptography algorithms include Advance Encryption Standard (AES), Data Encryption Standard (DES), 3Data Encryption Standard (3DES), Blowfish and secret key cryptography algorithms include Rivest–Shamir–Adleman (RSA) and Secure Force (SF) algorithm. The basic model of the cryptography is shown in fig.1.

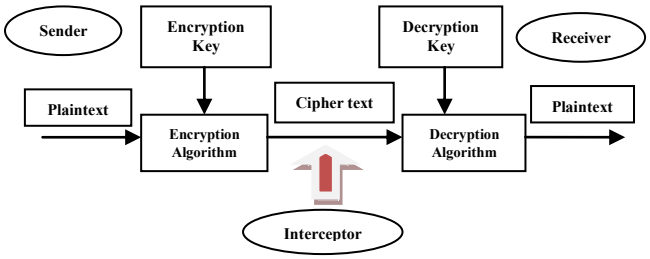

Fig.1: Basic Model of Cryptography

Digital watermarking

Digital watermarking is a process which keeps information safe. The basic model of digital watermarking is shown in fig.2. This process is to hide a set of owner's data within the materials, which should publish, with the purpose to be able to prove his ownership [3]. The watermarking of digital data has become very popular approach for intellectual property rights protection. The digital data such as text, image, audio, and video can be processed by the watermarking procedure. The requirements for watermarked data quality and safety for all types are also similar. The watermarking algorithm is consists of three steps - watermarking embedding, attacks and watermarking extraction. Application of digital watermarking includes medical application, copyright protection, finger printing etc. Digital watermark is classified into - visible watermark and invisible watermark [4]. Normally invisible watermark is better than visible watermark because in visible watermark, it is easy for the user to detect but difficult to detect automatically whereas in invisible watermark there is no major difference between the watermarked and the original image.

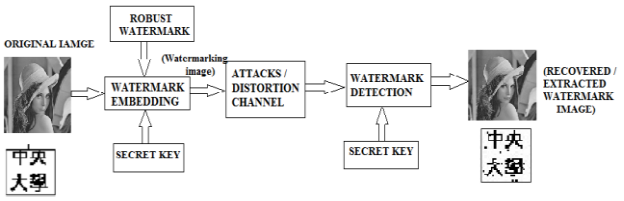

Fig.2: Basic Model of Digital Image Watermarking

Secure Force (SF) algorithm:

In Wireless Sensor Networks (WSN) the implementation of the SF algorithm offers very low complexity architecture [5]. The encryption and decryption process is almost similar in this algorithm. The five encryption rounds increases the effectiveness of the encryption procedure. This is a four step algorithm based on simple mathematical operations (six numbers of operations) that operates on a four bit data. But this causes a huge amount of misunderstanding and dispersal of data is created to combat dissimilar types of attacks. This algorithm consists of four steps [6]. Fig.3 shows the whole process of Secure Force algorithm step by step. The various steps of the SF Algorithms were studied in [7-8] which has been discussed in the following sections.

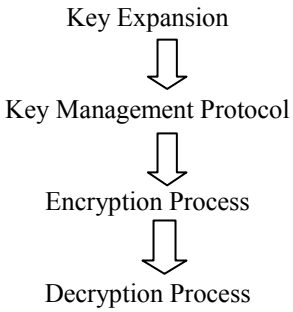

Fig.3: Secure Force Algorithm

Key Expansion: Different keys for encryption and decryption are produced in this process. This has reduced the likelihood of weak key and increased the key strength. This process is composed of two methods key expansion process and round key selection process [7]. Logical operation like XNOR and XOR, Left Shifting (LS), Permutation using P-table and Transposition using T-table are used in the key expansion technique [7]. The linear array of 64 bits is the cipher key. Cipher key is divided in four equal parts and each part consists of 2^4 bits. Every bit is organized in a 4X4 matrix horizontally and then shift-left operation is performed on this. The same technique is followed for vertical arrangement of the bits. A 64 bit linear array is formed by combining the output results. Further another shift left operation is performed on the final 64 bits and this is made to pass through permutation using P-Table and is placed in 4X4 matrix horizontally. This matrix is then multiplied with a pre-defined matrix or the FM (Fix Matrix) [7] which converts this 16 bits result into a 64 bits data. This converted 64 bits data are organized horizontally and the shift-left operation is again performed. Now AND & XOR operations are performed on the shift-left data which were divided into four vertically 16 bits blocks to move them into a unit 16 bit block. The resulting blocks are henceforth divided into 2^2 bits and are organized vertically to produce the 4 bit keys after the XOR operation is applied.

Key Management Protocol: For this step the Localized Encryption and Authentication Protocol (LEAP) is used which helps to send the key securely to the encoder [8]. This protocol is efficient in terms of energy and simplicity and mainly proposed for Wireless Sensor Network, which permits symmetric key establishment during the use of different key types. There are mainly four types of keys which are the group key, individual key, pair wise shared key and cluster key [8]. The LEAP key management protocol employs the pre-defined key to manage all the four types of keys for each and every node. A seed function and a node Identification (ID) are utilized to establish the individual. Next, in order to find the respective node IDs of the pair wise shared key, a neighbourhood detection procedure is used. The node at the receiver uses the seed function and initial key to calculate the shared key among all of its neighbouring nodes. Subsequently any intermediate and initial keys generated that are redundant are erased. After the setup of pair wise key, paired communication and paired secure key is done, it is used by the cluster head to distribute the cluster key. Ultimately the group key is transmitted by the Base Station.

Encryption process: The LEAP protocol creates a secure communication channel. The encoder receives the key securely and thus initiating the encryption process [8]. Logical operations like AND, XOR etc, substitution (S boxes), LS, and swapping are performed to create misunderstanding and dispersal in the encryption process. The 64 bits array of plain text is divided into two equal parts and then each of these parts are further broken down into two further equal parts. These small parts are swapped in each of the corresponding rounds. This is carried out to generate a more complicated cipher. Each round gives an output which is fed to the input of next round. It is mapped with the F-function [8]. Equation 1 explains the F-function mathematically.

$$F = OR(S - boxes\ (AND\ (LS\ (16\ bits/4)))) \quad (1)$$

The F-function pays a pivotal role in this process that induces the dispersal of data. It presents AND/ OR operations, S-boxes, and shift operations on the data. We perform XOR operation on the output from the F-function and the swapped data of the same round resulting in confusion of data. With this the encryption process is completed.

Decryption process: At the receiver end a reversal of the encryption method must be incorporated to receive the correct data. This technique is the Decryption Process.

II. RELATED WORK

A hybrid technique using Watermarking and Cryptography was presented in [9], for a transmission of protected text message. This scheme was based on XOR cipher, Fibonacci series, PN sequence, RSA, Hill cipher, one bit, two bit Least Significant Bit (LSB) and three bit LSB. They evaluated the quality of watermarked images on the basis of Root Mean Square Error (RMSE), Peak Signal to Noise Ratio (PSNR), and Mean Square Error (MSE). It was observed that the one-bit LSB watermarking was better as related to the two-bit LSB watermarking and three-bit LSB watermarking because MSE and RMSE were small and PSNR was large. They presented input message of variable length; the messages were encryption and decryption using the cryptography algorithms and the encrypted message hiding by three various technique of LSB watermarking.

The study of hybrid cryptographic encryption methods and also use of other encryption techniques to improve their level of security and also to study their amalgamation of hybrid techniques which was included the combination of cryptographic and digital watermarking methods was presented by Jai Singh, Kamil Hasan and Ravinder Kumar [10]. In terms of security Hybrid approach has was found to be more secure to hacking and illegal decryption of the information was impossible.

Use of Steganography for Image Security System was presented by Pooja Rani and Apoorva Arora [11] for some image security systems. Most of the prevailing image security systems are not up to date to defend against the least cyber attacks. In [11] MATLAB was used for the implementation and design of the system. Compression technique was used to reduce the size of the steganographed image. The original data (image) would be hidden into a different image. In the cluster based steganographic method the real and face image data are examined and wherever the color schemes of real and face image are alike, the real image would be implanted in those regions of the face image [11]. The main compression methods studied here include Discrete Cosine Transformation (DCT), Discrete Fourier Transformation (DFT), and Discrete Wavelet Transformation (DWT) because of the promising results obtained by their use in the field of cryptography (clustering). The time elapsed in the process is calculated. PSNR and MSE were also calculated for the different parameters to measure the image quality.

III. PROPOSED WORK

Here we are using Secure Force (SF) algorithm for the encryption and decryption process. For security purpose we are using watermarking on the image. For result and simulation we are using MATLAB software. Fig.4 shows the whole process.

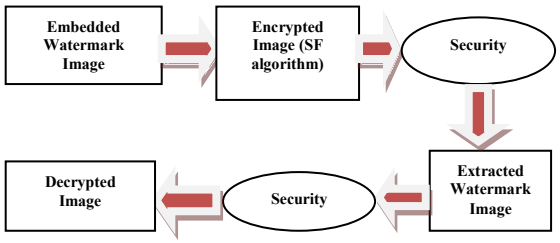

Fig.4: Process of the hybrid technique

Algorithm for hybrid digital watermarking and cryptography:-
Step 1- Read any Image file.
Step 2- Add watermark on the original image file.
Step 3- By using embedded technique we blur the original image.
Step 4- Convert the input image into grayscale image.
Step 5- By using key expansion and selection process input a specific key.
Steps 6- Now encrypt the blur image by using SF algorithm and get encrypted image, which may be noisy.
Step 7- The generated key applied to the image cipher.
Step 8- By using extraction technique gets the grayscale image.
Step 9- Set the hybrid level security system using password, OTP to open decrypt the image.

The above algorithm was implemented for on a standard MATLAB image.

IV. RESULT AND SIMULATION

In this paper we have used MATLAB software on the Standard image (Lena) data to implement our proposed technique. The SF-algorithm and watermarking process have been used to implement our hybrid technique. By the use of these two processes we have studied the decrypted image which was found to be a close match of the input image.Fig.5 shows the original image of Lena. Then we added watermarking on the original image. In the watermarking process we have included all the three steps of embedding, attacks, and extracting. Fig.6 shows the embedded image with the blur attack. Then we used SF-algorithm for encryption and decryption purpose. Fig.7 and 9 shows the encrypted and decrypted image of embedded image of Lena respectively.

Fig.5: Original Image of Lena

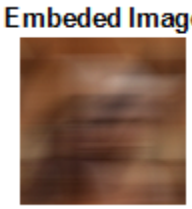

Fig.6: Embedded Image

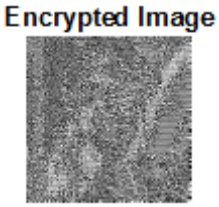

Fig.7: Encrypted Image

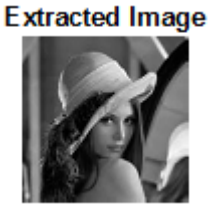

Fig.8: Extracted Image

Fig.9: Decrypted Image

Fig.8 shows the extracting image from the embedded image to decrypt the image and get best quality image.

Then we find the histogram of these images of Lena to study the quality of the decrypted image and how much variation have we got from the original one. The histogram of original image of Lena is shown in Fig.10. The histogram of encrypted image is shown in Fig.11. The histogram of decrypted image is shown in Fig.12.

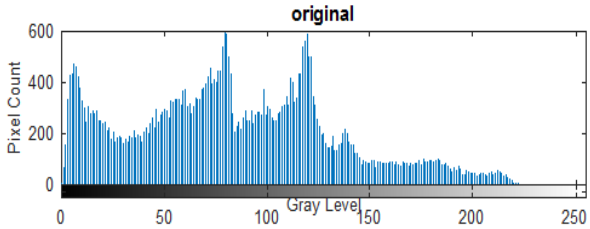

Fig.10: Histogram Plot of Original Image (Lena)

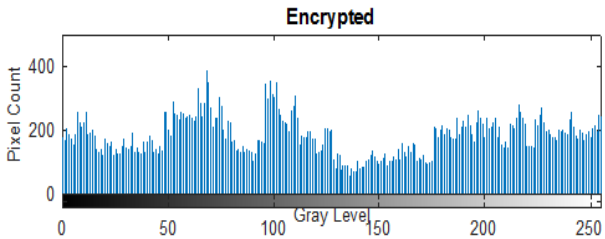

Fig.11: Histogram Plot of Encrypted Image (Lena)

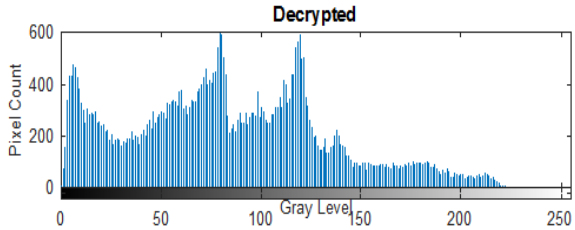

Fig.12: Histogram Plot of Decrypted Image (Lena)

The time elapsed for the complete encryption decryption process was calculated to be 114.332762 seconds and the MSE and PSNR were calculated by using the standard image of Lena and compared with a previous work.

TABLE 1: Comparision of PSNR, MSE

Parameter	PSNR	MSE
SF Algorithm [7]	10.7949	541.49
Proposed Hybrid Technique	34.4427	23.56

Table1 shows the calculation of PSNR and MSE of Lena image with a dimension of 220 X 220. We see that the value of MSE was better with respect to only SF Algorithm, but some optimization in the our proposed technique must be done to achieve a better PSNR. The entropy of the original image was found 7.6030 and that of the decrypted image was 7.6167. The percentage change of entropy was calculated to be 1.37%.

V. CONCLUSION

A hybrid technique for encryption and decryption of the image by using cryptography and watermarking process was implemented here. This was done in order to achieve higher level of Security. Level wise we can choose the level of security to secure the image safely without any corruption. Using this technique we can secure image, which can't be shared with anyone else. It gives the owner and authorized receiver of the image access to the data using different level of security. In this paper we have calculated the MSE which we saw was better that non hybrid technique using only SF algorithm. We did not achieve a better PSNR value and so some optimization our proposed technique must be done. The entropy of Lena image and the elapsed time for the complete process was also calculated. Further work need to be done to optimize and compare our work with other hybrid algorithms to achieve better results.

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